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A Project Report

On

PROJECT TITLE:

**"Codon-Helix: A DNA-Inspired Dual-Stream Triplet-Based
Compression Algorithm with Intrinsic Fault Tolerance"**

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For

Advanced Data Compression Techniques

Slot: E2 +TE2

Chapter 1 Abstract

Data integrity and efficient storage are critical requirements in modern computing systems, particularly in environments prone to noise, degradation, or transmission errors. Inspired by the biological mechanisms of DNA encoding and structure, this research proposes **Codon-Helix**, a novel data compression algorithm that combines **triplet-based codon encoding** with a **dual-stream redundancy model** mimicking the DNA double helix.

In this model, binary data is first grouped into **triplet codons**, similar to the three-base encoding of amino acids in DNA. These codons are assigned based on **frequency-aware probabilistic mappings**, providing a statistically optimized compression layer similar to Huffman or ANS coding. Simultaneously, the data is distributed across two logically intertwined streams—a **primary stream** and a **complementary stream**—designed using biologically inspired complementarity and parity rules. This dual-stream structure enables **built-in error detection and partial data recovery**, thereby offering both compression and fault tolerance within a unified framework.

The Codon-Helix model is evaluated against standard compression algorithms in terms of **compression ratio, resilience to corruption, decoding time, and recovery rate**. Experimental results demonstrate that the proposed approach achieves competitive compression while significantly improving data survivability under simulated fault conditions. This biologically inspired framework opens new pathways for the development of **resilient, low-complexity compression algorithms** suitable for critical systems, embedded devices, and low-bandwidth communications.

Chapter 2 Introduction

In the age of exponential data growth, the need for efficient, reliable, and resilient data compression methods has become increasingly critical. Conventional compression algorithms, such as Huffman coding, Lempel-Ziv (LZ) variants, or entropy-based models, have achieved notable success in optimizing storage and transmission. However, these methods typically focus solely on data compactness and often treat error resilience as a separate layer handled through error-correcting codes. This separation creates inefficiencies, particularly in **resource-constrained or high-risk environments** such as embedded systems, remote sensing, and space communication, where **data integrity and compression must coexist natively**.

In contrast, biological systems—most notably DNA—present a model of **ultra-dense information storage coupled with built-in redundancy and fault tolerance**. DNA's structure, characterized by the **double helix and codon-based encoding**, enables not only efficient information storage within living organisms but also **automatic replication, error checking, and repair**. This offers a compelling source of inspiration for designing novel data compression paradigms that merge compact representation with internal fault resilience.

This paper introduces **Codon-Helix**, a DNA-inspired data compression framework that synthesizes two biological concepts: (i) **codon-based encoding**, in which data is grouped and encoded in fixed-length triplets, and (ii) a **helix-based dual-stream structure**, simulating the

redundancy of DNA's antiparallel strands. The proposed model maps binary data into symbolic codons using frequency-aware encoding, then replicates the stream into a secondary complementary strand, enabling partial recovery in case of corruption or data loss.

The **primary objectives** of this research are:

- To design a **compression model** that integrates triplet-based codon encoding with a biologically motivated redundancy scheme.
- To achieve a **favorable balance between compression efficiency and fault tolerance**.
- To evaluate the system's performance under various corruption scenarios and benchmark it against existing compression standards.